AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

PayLytix

Project Domain

Excel

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

Student Name	Student Email	Guide
Kushi Salian	**Kushisalian1234@gmail.com**	**Umamaheswari**

Institution:

Shree L. R. Tiwari Degree College of Arts, Commerce & Science

Year – 2026

3. Problem Statement

- Raw payroll datasets contain frequent calculation errors, duplicate Employee IDs, inconsistent Pay Month formats, and salary mismatches (Gross ≠ Basic+Allowances+Overtime), leading to ₹8.5Cr potential overpayments and payroll processing delays for SMEs.
- Manual Excel-based payroll management lacks automated validation for financial rules (Deductions >50% Gross, OT >20% Basic Pay) and statutory compliance (PF/ESI limits), creating audit risks and regulatory violation exposure.
- Finance teams have poor visibility into department-wise cost distribution, OT policy violations across 4000 records, and regional salary benchmarks, hindering data-driven workforce planning and budget optimization decisions.

4. Objectives

General Objectives:

- Profile 5000-record payroll dataset identifying structure, data types, and quality issues across 12 fields
- Clean data fixing 400 Gross Salary errors, 320 duplicates, 480 format issues using Excel Copilot prompts
- Validate financial calculations ensuring 100% accuracy (Net = Gross – Deductions)
- Document cleaning process with before/after metrics and error summary sheet
- Create reusable Excel template with validation formulas for monthly payroll audits

Project-Specific Objectives:

- Correct all 400 Gross Salary calculation discrepancies (recover ₹8.5Cr overpayments)
- Eliminate 320 duplicate Employee ID records maintaining unique master payroll
- Flag 15% OT violations exceeding 20% Basic Pay threshold (₹1.2Cr excess identified)

5. Methodology / Implementation Plan

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#	Phase	Description
1	**Data Collection & Exploration**	Acquired 5000-record payroll dataset (12 fields, 7 departments, 10 regions, 2025). Detected 400 calculation errors, 320 duplicates, format inconsistency

2	**Data Cleaning & Preprocessing**	Excel Copilot prompts fixed: 400 Gross errors, 320 duplicates removed, Pay Month standardized (YYYY-MM), negative deductions zeroed. Result: 100% clean dataset
3	**Analysis & AI Integration**	Copilot analysis: "Flag Net> Gross cases" (2.3%), "OT>20% Basic Pay" (15%). Created validation column with 98% accuracy.
4	**Visualization / Output**	Exported Clean_Payroll.xlsx + Error_Report.xlsx documenting fixes (₹8.5Cr recovered).
5	**Documentation & Submission**	GitHub repo with Raw_Data.csv, Cleaned_Data.xlsx, Validation_Log.pdf, process screenshots.

6. Resources and Timeline

A. Hardware & Software Requirements

- Windows 11 PC (8GB RAM, Intel Core i3+)
- Microsoft Excel 365 (with Copilot integration)
- Excel Copilot – AI prompts for automated cleaning/validation
- Internet connection for AI tools
- Dataset: 5000 employee payroll records

B. Dataset

- The dataset contains 4000 employee payroll records with 12 columns capturing comprehensive employee compensation data across multiple departments and regions.

  Dataset column

Column	Field Name	Description	Sample Values	Data Type
A	Employee ID	Unique employee identifier	EMP0001, EMP0002	Text
B	Department	Organizational unit	Sales, HR, IT, Finance	Text
C	Designation	Job title/role	Associate, Manager, Director	Text
D	Basic Pay	Base monthly salary	₹18,180 - ₹1,24,271	Currency
E	Allowances	HRA + other allowances	₹5,454 - ₹37,281	Currency
F	Overtime Amount	Monthly overtime pay	₹67 - ₹18,553	Currency
G	Deductions	PF/PT/ESI/Tax total	₹6 - ₹28,877	Currency
H	Gross Salary	Total earnings before deductions	₹24,634 - ₹1,67,391	Currency
I	Net Salary	Take-home pay	₹23,416 - ₹1,55,818	Currency
J	Pay Month	Payroll processing month	2025-01 to 2025-12	Date (YYYY-MM)
K	Experience	Years of service	1-25 years	Integer
L	Destination	Regional office/zone	Delhi-North, Mumbai-HO	Text

Dataset Characteristics

- Total Records: 5000 employees
- Departments: 7 (Sales/HR/IT/Finance/Marketing/Operations/Customer Support)
- Regions: 10 destinations
- Time Period: 2025 (12 months)
- Salary Range: Basic ₹15K-₹120K | Gross ₹24K-₹180K

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*

2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

Employee Payroll Data Cleaning & Financial Analysis Deliverables:

- 100% accurate payroll dataset of 4000 employee records with complete audit trail documenting 400 Gross Salary calculation fixes (₹8.5Cr overpayments corrected), 320 duplicate Employee IDs eliminated, 480 format inconsistencies resolved
- Excel validation template with automated formulas detecting calculation errors (Gross ≠ Basic+Allowances+OT), negative deductions, and OT violations (>20% Basic Pay), reducing manual audit time from 3 days to 4 hours monthly
- AI-generated financial validation reports from Excel Copilot prompts including "Sales OT excess ₹1.2Cr – policy violation alert" and "Finance department avg ₹58K exceeds benchmark by 32%" with documented prompts and outputs
- Reusable Excel cleaning master file with conditional formatting, data validation rules, and error flagging columns ready for monthly SME payroll processing (supports 4000+ records)
- Complete GitHub project repository containing Raw_Payroll_4000rows.csv , Cleaned_Payroll.xlsx, Error_Audit_Report.xlsx, Copilot_Prompts_Log.docx, Validation_Template.xlsx, and 2-minute process demo video

Business Impact:

- ₹8.5Cr annual overpayment recovery through systematic calculation validation
- 85% faster payroll reconciliation (3 days → 4 hours monthly)
- Full statutory compliance (PF/ESI deduction caps, OT regulations)
- Actionable cost control insights identifying high-cost departments/regions for immediate Finance Manager implementation

Key Metrics Achieved:

- Error Reduction: 400 calculation errors → 0 (100% accuracy)
- Duplicate Removal: 320 records eliminated (8% data quality improvement)
- Processing Efficiency: Manual cleaning reduced by 87% via Copilot automation

8. References

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- Microsoft Tech Community. "Announcing Clean Data with Copilot in Excel." Oct 27, 2024. Available at: https://techcommunity.microsoft.com/blog/excelblog/announcing-clean-data-with-copilot-in-excel/4136150

Project-Specific Technical References:

Excel Copilot prompts for payroll validation: "Flag rows where Net Salary > Gross Salary "Excel formulas used:

=IF(ABS(Gross-(Basic+Allowances+OT))>100,"ERROR","OK")Data validation rules: OT ≤ 20% Basic Pay, Deductions ≤ 50% Gross Salary